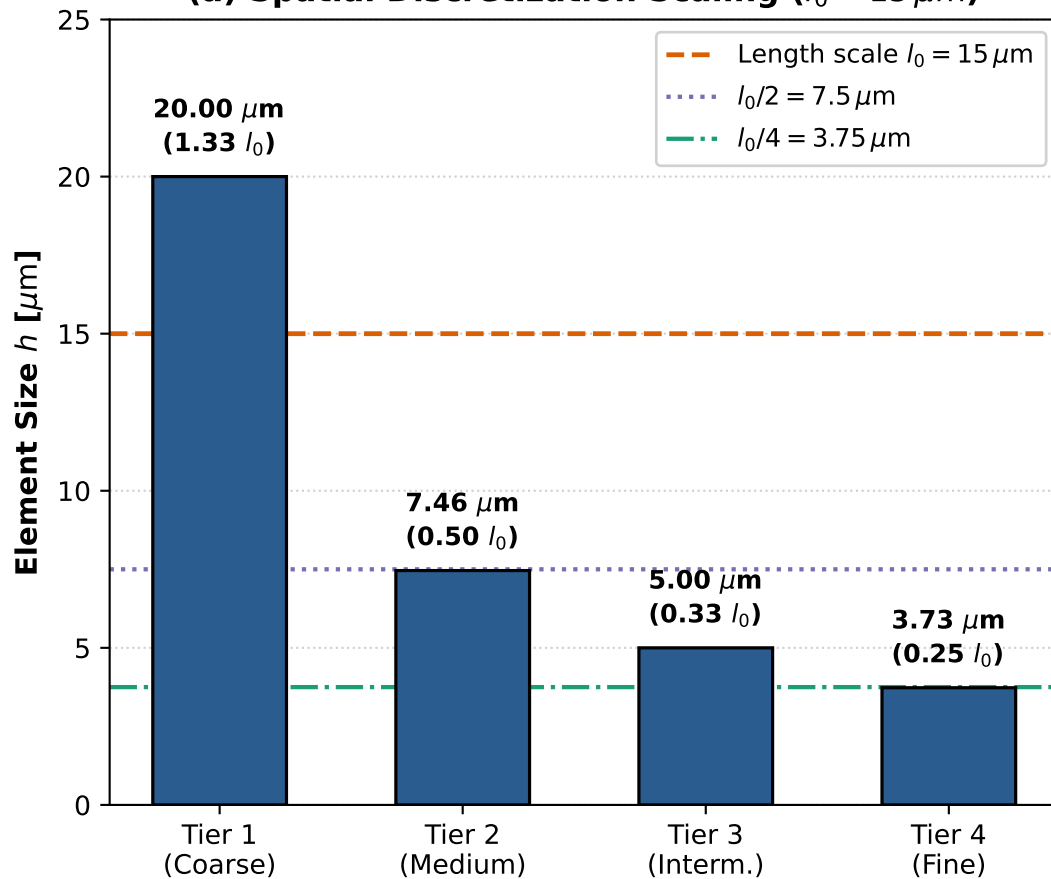
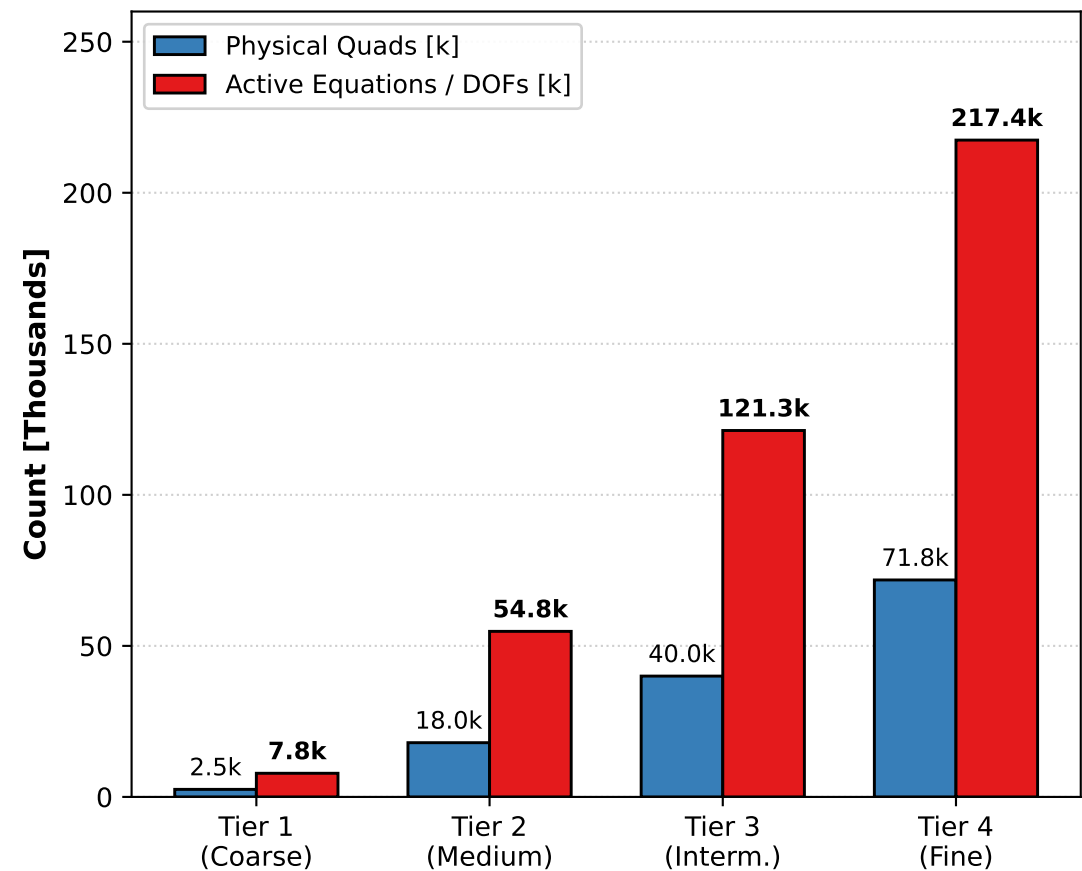


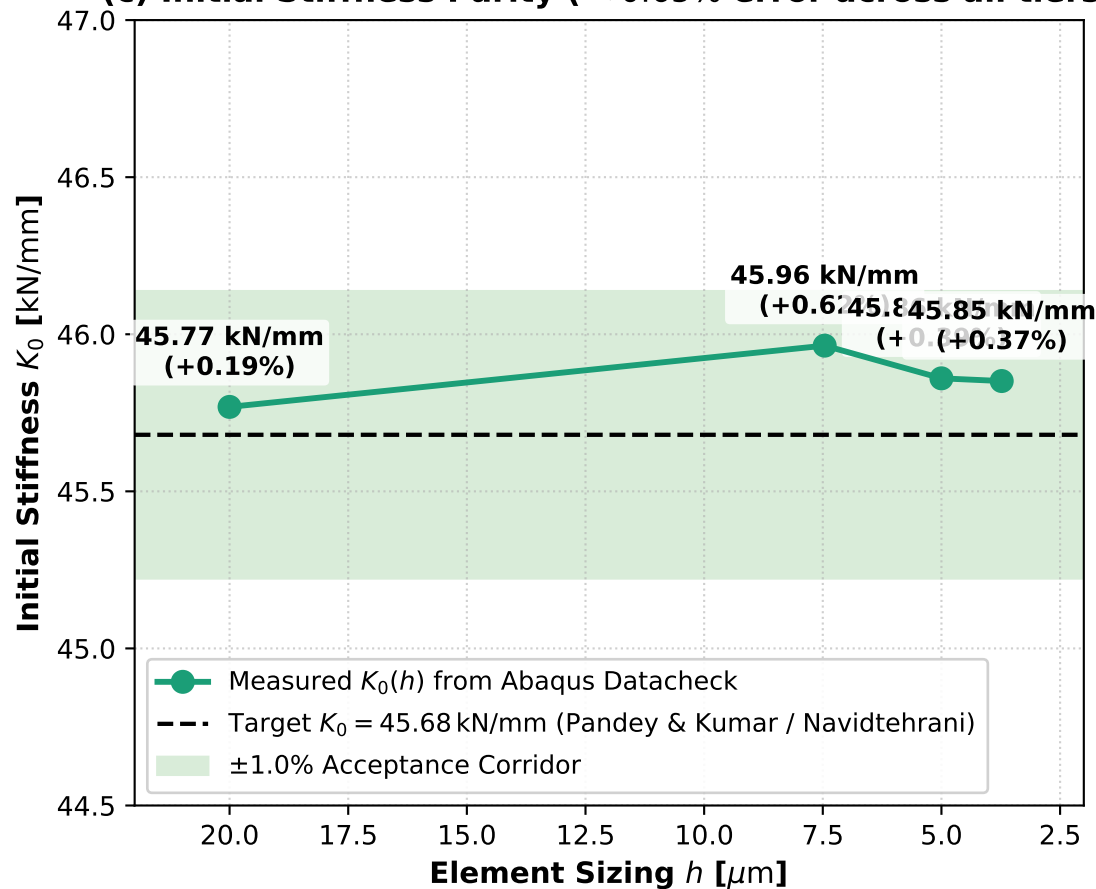
(a) Spatial Discretization Scaling ($l_0 = 15\ \mu\text{m}$)



(b) Computational Problem Size (2.5k $\rightarrow$ 71.8k FEs)



(c) Initial Stiffness Parity ($< 0.65\%$ error across all tiers)



(d) Epistemological Falsification Matrix

GATE M2-1B: FIXED-MESH SPATIAL CONVERGENCE
Boundary Value Problem: $u_y = 0$, $E = 210\ \text{GPa}$, $\nu = 0.3$, $G_c = 2.7\ \text{N/mm}$, $l_0 = 15\ \mu\text{m}$

As $h \rightarrow 0$ ($20 \rightarrow 3.73\ \mu\text{m}$), does $F_{\max}(h)$ converge to:

POSSIBILITY A:
SOLVER CONCURRENCE
($F_{\max} \rightarrow 410\text{--}415\ \text{N}$)

- Fixed mesh reproduces $\sim 412\ \text{N}$
- Proves adaptive remeshing is accurate
- Literature discrepancy is external
- Mathematical formulation verified

POSSIBILITY B:
ADAPTIVE FAILURE
($F_{\max} \rightarrow 360\text{--}370\ \text{N}$)

- Fixed mesh converges to $\sim 365\ \text{N}$
- Proves adaptive remeshing failed
- Refinement corridor too narrow
- Over-stiffening artifact isolated